

Bernice Mercy Sharon Malaiarasu

Boston, MA | (617) 792-6874 | Gmail | LinkedIn | Github | Portfolio

Profile

Generative AI engineer with 4+ years of software/data engineering experience, including 2+ years building production LLM systems at Thoughtworks. Recipient of the Bayer GenAI Award for Best Technical Implementation and acknowledged contributor on the resulting Frontiers in AI paper. Pursuing an MS in AI at Northeastern's Khoury College; seeking a Spring 2027 Agentic AI internship.

Experience

AIMES Lab, Northeastern University <i>Research Assistant</i>	Boston, MA <i>Jul 2026 – Present</i>
• NewsroomFeed – AI Journalism Platform – LangGraph pipeline turning six live civic feeds (MassDOT, TomTom, NWS, Boston 311) into hyperlocal AI news, with journalist approval before publish; newsworthiness scored as severity × likelihood per ISO 31000. Shipped as a Next.js PWA over a FastAPI service with geolocated feed, commute-route push alerts, and an editorial review queue. • Husky AI – Interactive platform for students to practice and refine prompt engineering with guided feedback.	
Thoughtworks <i>Sr. Consultant – Data Engineer</i>	Chennai, India <i>Aug 2022 – Dec 2025</i>
<i>PRINCE Chatbot – Generative AI</i>	<i>Jan 2024 – Dec 2025</i>
• Led development of PRINCE (LangChain, LangGraph) — a supervisor-orchestrated multi-agent system with tool calling, deployed in production at Bayer across 18,000+ preclinical safety studies, cutting query response time by 30% and automating report generation. • Built the retrieval layer: 100 GB+ of biomedical data indexed in OpenSearch (vector store) with hybrid semantic + keyword search, 5x query expansion and cross-encoder re-ranking, every response citation-linked to source; plus Text-to-SQL on Claude 3.5 Sonnet using few-shot prompt engineering and structured output validation, pushing SQL accuracy above 90% against AWS Athena and cutting ad-hoc data requests from days to minutes. • Owned output quality: NER pipeline at 98%+ accuracy across 40+ entity fields (data correction from days to 15 minutes, surfacing entry errors in 30%+ of records missed in prior manual review), and RAGAS + DeepEval as a CI/CD gate for MLOps auto-running expert-curated test sets across 5 metrics on every code or prompt change, with Langfuse observability and score drops triggering root-cause analysis before production.	
<i>PRINCE Data Platform – Data Engineering</i>	<i>Aug 2022 – Dec 2023</i>
• Processed 20,000+ preclinical study PDFs — extracting, chunking, and generating text embeddings into OpenSearch as the vector store powering PRINCE's RAG retrieval. • Migrated 56 database tables from Kubernetes to AWS with Terraform-managed pipelines syncing into Elasticsearch as the query layer for the preclinical research API; built and maintained daily Spark (Scala) + AWS Glue ETL across 70+ source and 20+ sink tables.	

Projects

Entiscribe – PDF Entity Extraction & Knowledge Graph	July 2026
• LangGraph pipeline (ingest, extract, evaluate, knowledge-graph assembly) extracting user-defined entity types from PDFs; scores every extraction with gold-standard precision/recall/F1 against a reference set plus an always-on LLM groundedness judge, canonicalizes entities that co-refer across files, and renders an interactive knowledge graph. Runs fully local with 0 API keys.	
Job Applier – AI-Assisted Job Search Platform	July 2026
• Resume and cover-letter tailoring against real job descriptions with a no-fabrication guardrail — never invents an employer, date, skill, or metric absent from the master resume — and deterministic keyword-classified fit scoring rather than a drifting one-shot LLM judgment; browser automation stops at the final human-confirmed click. Flask app + Manifest V3 Chrome extension.	
Self-Harm Prevention Classifier <i>Northeastern</i>	June 2026
• RoBERTa-base multi-label crisis classifier on CRADLEBench (1,308+ clinician-annotated cases, 7 categories): 0.785 macro F1, 0.920 flagged-F1 — within ~2.7 pts of GPT-5 at 100–500× smaller. QLoRA fine-tune of Llama-3.2-1B cut completion loss 35% and perplexity 67% over base.	
E-Commerce Delivery Delay Prediction <i>Northeastern</i>	Apr 2026
• Two-stage XGBoost delay predictor on 100K+ OLIST orders — a classifier flags risk, a regressor trained only on late orders estimates magnitude: 0.922 ROC-AUC against a 6.8% real-world delay base rate. Used SHAP to identify the features actually driving delays.	

Education

Northeastern University – Khoury College of Computer Sciences	Boston, MA
<i>MS in Artificial Intelligence, Specialization in Machine Learning</i>	<i>Jan. 2026 – May 2028</i>
Loyola ICAM College of Engineering and Technology	Chennai, India
<i>BE, Electronics and Communications Engineering</i>	<i>Aug. 2018 – May 2022</i>

Awards & Talks

Bayer GenAI Award Best Technical Implementation for multi-agent GenAI system at Bayer	2025
Frontiers in AI Acknowledged contributor, PRINCE knowledge engine (DOI: 10.3389/frai.2025.1636809)	2025
Speaking XConf & Geeknight (PRINCE architecture, LLM eval, Text-to-SQL); GenAI/RAG workshop, 50+ devs	2024–25

Technical Skills

AI/ML: LLMs, RAG Pipelines, Agentic AI, Multi-agent Systems, LangChain, LangGraph, Langfuse, RAGAS, DeepEval, Tool Calling	
Cloud: AWS (Glue, Athena, OpenSearch, DynamoDB, S3, Step Functions)	
Languages & Frameworks: Python, Scala, SQL, PyTorch, TensorFlow, HuggingFace, OpenCV, Databricks/Spark, Terraform, Docker, Git	